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Stochastic Network Resilience Under Random Failures

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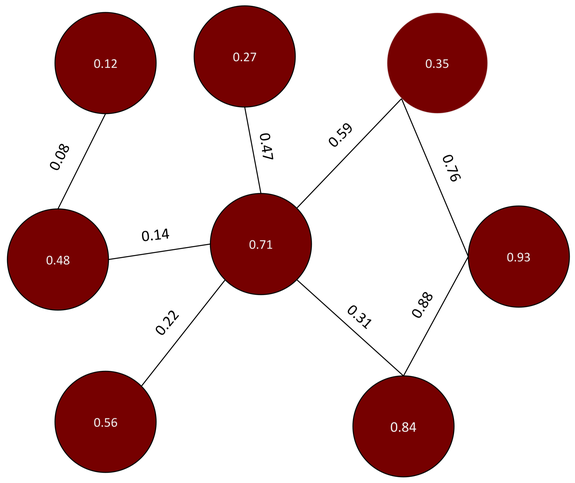
Dept. of Mathematical Sciences and Systems Engineering, Florida Institute of Technology

Introduction:

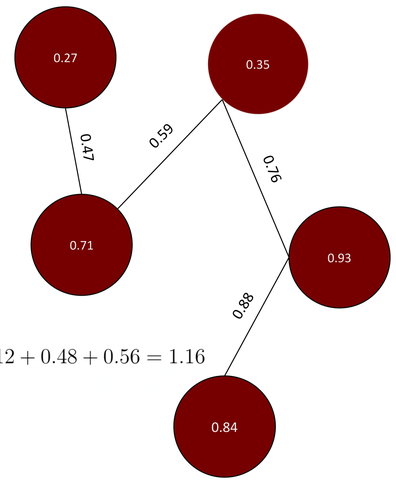
This project investigates threshold-crossing probabilities in stochastic networks. Using probabilistic modeling and transform-based analytical methods, the work derives expressions that characterize when cumulative losses exceed prescribed limits with the goal of providing insight into the relationship between local random behavior and global network risk.

Diagram:

Consider the graph:



After an attack that same graph becomes:



We have X_3^j:

$$X_3^j = W_e + W_n$$

Where W_n is:

$$W_n = \sum_{k=0}^{X_1^j} Y_k = Y_0 + Y_1 + Y_2 = 0.12 + 0.48 + 0.56 = 1.16$$

and W_e is:

$$W_e = \sum_{k=0}^{X_2^j} Z_k = Z_0 + Z_1 + Z_2 + Z_3 = 0.08 + 0.31 + 0.22 + 0.14 = 0.75$$

Theorem 1:

Let $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_2$ denote the first crossing index (of X_1^j, X_2^j, and X_3^j respectively); and thresholds $\mathbf{M}_1$, $\mathbf{M}_2$, and $\mathbf{M}_3$. Then, for $P(\mathbf{v}_1 < \mathbf{v}_2 < \mathbf{v}_2)$, from White **[1]** we have:

$$P(\nu_1 < \nu_2 < \nu_3) = H_x^{-1}\left(\frac{\gamma(1,x_2,x_3)-\gamma(x_1,x_2,x_3)}{1-\gamma(x_1,x_2,x_3)} \cdot \frac{\gamma(1,1,x_3)-\gamma(1,x_2,x_3)}{1-\gamma(1,x_2,x_3)}\right)(M),$$

Where:

$$H^{-1} = LC_\omega^{-1} \circ D_\varepsilon^{M_2-1} \circ D_\nu^{M_1-1}, \qquad M = (M_1, M_2, M_3) \text{ is the threshold vector.}$$

And:

$$\gamma(\eta, \varepsilon, -\ln(w)) = \mathbb{E}\left[\eta^{X_1^j} \varepsilon^{X_2^j} w^{X_3^j}\right] = g_1\left(\eta\, g_2(\varepsilon\, \ell_3(w))\right),$$

Theorem 2:

We can find the closed form probability of a given threshold crossing given a set distribution of X_1^j, X_2^j, and X_3^j.

Letting: $X_1^j, X_2^j \sim \text{Geom}, \quad$ and $\quad X_3^j \sim \text{Exp}(r),$

We find: $P(\nu_1 < \nu_2 < \nu_3) =$

$$(S_N)\left\{p_2^{M_1-1}\left[P(M_2-1, rM_3) - \frac{e^{-rp_1p_2M_3}}{(1-p_1p_2)^{M_2-1}} P\big(M_2-1, (1-p_1p_2)rM_3\big)\right]\right\}.$$

From the distribution of X_1^j, X_2^j, and X_3^j we get:

$$g(z) = \frac{pz}{1-qz}, \qquad \text{and} \qquad \ell_3(z) = \frac{r}{r+z}.$$

Then:

$$P(\nu_1 < \nu_2 < \nu_3) = \left(LC_\omega^{-1} \circ D_\varepsilon^{M_2-1} \circ D_\nu^{M_1-1}\right)\left(F(\nu,\varepsilon,\omega)\right)(M)$$

$$\text{(i)} = \left(LC_\omega^{-1} \circ D_\varepsilon^{M_2-1}\right)\left[K(\varepsilon,\omega)\big(g_2(\varepsilon\ell_3(\omega))\big)^{M_1-1}\right]$$

$$\text{(ii)} = LC_\omega^{-1}\left[K(\omega)\sum_{t=0}^{N}\binom{M_1+t+2}{t} C^t\left(\frac{B^{N+t+1}-A^{N+t+1}}{B-A}\right)\right]$$

$$\text{(iii)} = (S_N)\left\{p_2^{M_1-1}\left[P(M_2-1, rM_3) - \frac{e^{-rp_1p_2M_3}}{(1-p_1p_2)^{M_2-1}} P\big(M_2-1, (1-p_1p_2)rM_3\big)\right]\right\}$$

$$\therefore \quad P(\nu_1 < \nu_2 < \nu_3) =$$

$$(S_N)\left\{p_2^{M_1-1}\left[P(M_2-1, rM_3) - \frac{e^{-rp_1p_2M_3}}{(1-p_1p_2)^{M_2-1}} P\big(M_2-1, (1-p_1p_2)rM_3\big)\right]\right\}.$$

Here p_1, q_1, p_2, q_2, and r are the distribution parameters of X_1^j, X_2^j, and X_3^j; M_1, $M2$, *and* $M3$ are the crossing thresholds of their respective components; $P(a,z)$ is the lower regularized incomplete gamma function; and finally, where we define:

$$K(\varepsilon,\omega) = \frac{p_1^2 p_2^2 (1-\varepsilon)\varepsilon\, \ell_3(\omega)^2}{(1-A\varepsilon)(1-A)(1-\varepsilon\ell_3(\omega))},$$

$$K(\omega) = \frac{p_1^2\big(p_2\ell_3(\omega)\big)^{M_1+1}}{1-A},$$

$$A = (1-p_1p_2)\ell_3(\omega), \qquad B = \ell_3(\omega), \qquad C = q_2\ell_3(\omega),$$

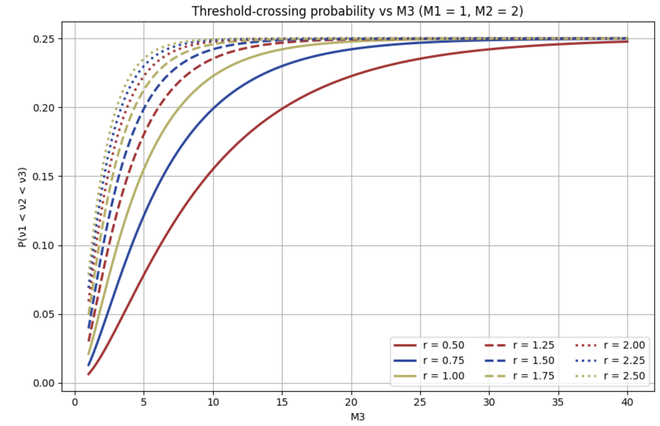
$$S_N = \sum_{t=0}^{N}\binom{M_1+t+2}{t} q_2^t\left[1-(1-p_1p_2)^{N+t+1}\right],$$

$$N = M_2 - M_1 - 1.$$

and :

$$F(\nu,\varepsilon,\omega) = \gamma(\eta,\varepsilon,-\ln(w)) = g_1\left(\eta\, g_2(\varepsilon\, \ell_3(w))\right),$$

Example:



References:

[1] R. T. White (2022). On the Exiting Patterns of Multivariate Renewal-Reward Processes with an Application to Stochastic Networks. *Symmetry*, 14(6), 1167.

[2] J. H. Dshalalow (1997). On the Level Crossing of Multidimensional Delayed Renewal Processes. *Journal of Applied Mathematics and Stochastic Analysis*, 10(4), 355-361.